Ronith Kumar Devarakonda

ronithkumardevarakonda@gmail.com Ronith Kumar Devarakonda Portfolio

Education

Expected in Aug, 2024 — **Bachelor of Technology** Indian Institute of Technology Delhi
Electrical Engineering (Major), CGPA: 7.03

Bachelor of Technology Indian Institute of Technology Delhi
Computer Science and Engineering (Minor)

Apr, 2020 — **Board of Intermediate Education** Sri Chaitanya Jr. College, CGPA: 9.82

Apr, 2018 — **Board of Secondary Education** Dr.K.K.R's Gowtham School, CGPA: 9.80

Scholastic Achievements

2023 — **IELTS**, 7.5(Good User)

2020 — **Joint Entrance Examination (JEE) Advanced**, 28(ST) rank

Joint Entrance Examination (JEE) Mains, 25(ST) rank

National Aptitude Test in Architecture (NATA), 9(ST) rank

Qualified for **KVPY Fellowship Award**

KVPY Aptitude Test, 116(SX stream) rank

TS Eamcet, 1393 rank

Projects

B.TECH Project:

2023 — **Wireless Power Transfer, Prof. Sumit Kumar Pramanick**

- Designed a **wireless power transfer system** for a 40W bulb in Inductive power transfer method and worked on optimizing the coils used for power transfer by varying different parameters like no. of turns, width of the coil in order to minimize the power losses
- Designed the system using inverters, rectifiers and inductor coils and then designed the **coil parameters** of the inductor using appropriate calculations
- Simulated the entire system in **MATLAB simulink** and got appropriate results. We collaborated with other team for optimized capacitors to get the appropriate results by conducting experiment manually

Current Projects:

2024 — **Unity Game Developer, Prof. Prem Kumar Kalra**

- Railway Men is a search and rescue theme-based game. It is a single player game. Players need to search and rescue people from a small-medium scale length railway station
- They need to clear and move from different objectives to finish the game. Games assets are imported from licensed assets and some of them are created in Blender. Using these assets text-based game is programmed in Unity

Implementation of computer graphics techniques, Prof. Rahul Narain

- Getting familiar with foundations and rasterization by creating a clock and 3d scene using software and hardware rasterizer
- Implementing modelling by using half edge data structure to create, load and connect the meshes
- Path Tracing , Simulation and key framing are yet to be done

Projects (continued)

Completed Projects:

2023 **Performing various image processing techniques using open-cv library, Prof. P.K.Kalra**

- **Artistic Image Enhancement and Style Transfer**, Rendering of real photos into non-photo realistic painting like images using the light and shadow relation
- **Facial Expression Transfer and Face Swapping**, Face expression transfer using barycentric co-ordinates and face swapping using face warping. **Seam Carving**, Re-sizing the images using vertical or horizontal seams (which have minimum energy) and training a learning model

Frequent Itemset Mining with FP-Growth Algorithm, Prof. Sayan Ranu

- Implemented the FP-Growth algorithm to condense extensive datasets by recognizing frequent itemsets using the FP-tree structure. This method effectively reduces data dimensions, preserves vital patterns, and enhances storage efficiency

2022 **House Price Prediction, Prof. Brejesh Lall**

- Built a **3-layered neural network** using the **Keras** package to predict if house prices are above or below median value. Optimized the model using the **stochastic gradient descent** with momentum and used the **binary cross entropy** loss function. Employed grid search for optimising hyperparameters and achieved 95 percent test accuracy on Zillow's home value prices dataset

Classification of Hand written numbers into Digital numbers, Prof. Brejesh Lall

- Developed a machine learning model to categorise handwritten numbers into digital numbers $0-9$ using multivariate **logistic regression** using the MNIST dataset and the fundamental libraries **NumPy, Pandas, and Matplotlib**

2021 **DS Coin Crypto Currency Model, Prof. Amitabha Bagchi and Prof. V.V Koppula**

- Designed a project which makes all the cryptocurrency transactions smoothly. Every participant can be either seller, buyer, or miner(who verifies and approves the transactions). This project was implemented using different data structures like Merkle Trees, authenticated linked lists. Each transaction will be stored in Transaction blocks. All the transaction blocks are stored in the blockchain. For authentication purposes, the **SHA-256** algorithm was used. This project also handles some malicious activities. Using of Tree data structure makes this algorithm time efficient

Courses Completed

Relevant Courses:

- Intro. To Computer Science, **Data Structures and Algorithms**, Probability and Stochastic Processes, Discrete Mathematics, **Advanced Data Structures and Algorithms**, **Machine Intelligence and Learning**, **Operating Systems**, Data Mining, Digital Image Processing, **Computer Graphics**

Skills

Languages: C, **C++**, C Sharp, Python, Java, HTML, CSS, JavaScript, LaTeX

Software: Blender, Unity, Unreal Engine 5, Matlab, LT spice, Microsoft Office, Linux, Adobe Photoshop

Extra Curricular Activities

Clubs: I am the Team Supervisor in the **Aero-Modelling Club**, IIT Delhi. I am a cadet of NCC, IIT Delhi

Activities: Active participant in **Sports** and **e-Sports**, Active participant in competitions conducted by **Art Station**, **Unity** and **Blender**

Interests: Reading Novels specific to **Fiction** and **Thriller** genre, Watching **Anime** and **TV shows**